

Supplementary Material: C-STGB: Risk-Controlled Spatio-Temporal Graph Learning for Anti-Money Laundering under Extreme Imbalance and Topological Camouflage

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APPENDIX A COMPLIANCE RULE VERIFIER AST GRAMMAR & SCHEMA SPECIFICATION

To guarantee regulatory determinism and enforce zero entity hallucinations during automated Suspicious Activity Report (SAR) generation (Section V), the multi-agent forensic swarm employs a formal Abstract Syntax Tree (AST) grammar and runtime Pydantic schema validator.

A. Extended Backus-Naur Form (EBNF) Grammar

All natural-language and structured FinCEN Form 111 XML narratives generated by the SAR Drafter Agent must strictly conform to the following formal grammar $\mathcal{G}_{\text{SAR}}$:

```
SAR_Document ::= Header
                SubjectSection
                ActivitySection
                EvidenceSection
                AuditTrailer
Header ::= "<SAR_Form111"
        " Version='2.0'>"
        FilingInst
        FilingTimestamp
        CaseID
        "</SAR_Form111>"
SubjectSection ::= "<Subjects>"
                Subject+
                "</Subjects>"
Subject ::= "<Subject ID='"
            EntityID "'>"
            IdentityType
            RiskScore
            ConformalTier
            Jurisdiction
            "</Subject>"
ActivitySec ::= "<Activity>"
                Typology
                Amount TemporalSpan
                Narrative
                "</Activity>"
Typology ::= "Smurfing"
            | "CircularWash"
            | "RapidLayering"
            | "DormantPass"
            | "HubCamouflage"
EvidenceSec ::= "<Evidence>"
                Root CausalHop+
                FlowRatio Hawkes
```

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```
</Evidence>"
CausalHop ::= "<Hop Seq='" Int "'>"
            Src "-->" Dst
            Amt Time Gate
            "</Hop>"
FlowRatio ::= "<Flow In='" Real
            "' Out='" Real
            "' Ratio='" Real "/>"
AuditTrailer ::= "<Fed_SR11_7>"
                MerkleRoot
                SHA256Proof Time
                "</Fed_SR11_7>"
EntityID ::= [a-zA-Z0-9_]{1,64}
Real ::= [0-9]+ "." [0-9]+
Int ::= [0-9]+
```

B. Pydantic Runtime Schema Validation Specification

Before any FinCEN Form 111 XML payload is accepted into the Federal Reserve SR 11-7 audit repository or dispatched to human compliance officers (Tier 2), it is compiled into an AST and validated against the Pydantic schema $\mathcal{S}_{\text{AML}}$:

```
class CausalHopSchema(BaseModel):
    hop_seq: int = Field(
        ..., ge=1, le=10
    )
    source_entity_id: str = Field(
        ..., min_length=1
    )
    target_entity_id: str = Field(
        ..., min_length=1
    )
    amount_usd: float = Field(
        ..., gt=0.0
    )
    timestamp_iso: str
    edge_trust_gate: float = Field(
        ..., ge=0.0, le=1.0
    )

class SARDocumentSchema(BaseModel):
    case_id: str = Field(
        ..., regex=r"SAR-[0-9]{4}-[A-Z0-9]{8}$"
    )
    filing_timestamp: datetime
    conformal_tier: Literal[
        "Tier_1_Quarantine",
        "Tier_2_Abstention"
    ]
    predicted_risk_score: float = (
        Field(..., ge=0.0, le=1.0)
    )
    primary_typology: Literal[
        "Smurfing", "CircularWash",
        "RapidLayering", "DormantPass",
        "HubCamouflage"
    ]
    flow_conservation_ratio: float
    causal_trajectory: (
        List[CausalHopSchema]
    ) = Field(..., min_items=1)
    merkle_root_hash: str = Field(
```

```

..., min_length=64,
max_length=64
)

@validator("causal_trajectory")
def verify_topological_chain(
    cls, v
):
    for i in range(len(v) - 1):
        assert (
            v[i].target_entity_id
            == v[i+1].source_entity_id
        ), "Disconnected chain."
    return v

```

C. Unsupported-Entity Prevention Invariant & Runtime Compiler Specification

Proposition 1 (Deterministic Unsupported-Entity Prevention Invariant (Proposition S1)). *Let $\mathcal{G}_{sub} = (\mathcal{V}_{sub}, \mathcal{E}_{sub})$ denote the factual transaction subgraph extracted from the immutable ledger, and let $\mathcal{T}(\text{SAR})$ denote the parsed AST generated by the LLM-based SAR Drafter Agent. If $\mathcal{T}(\text{SAR})$ passes schema validation $\mathcal{S}_{AML}$, then:*

$$\forall u \in \mathcal{V}(\mathcal{T}), u \in \mathcal{V}_{sub} \quad \text{and} \quad \forall (u, v) \in \mathcal{E}(\mathcal{T}), (u, v) \in \mathcal{E}_{sub} \quad (1)$$

Proof. The Pydantic schema validator executes a deterministic set-membership assertion $\mathcal{V}(\mathcal{T}) \subseteq \mathcal{V}_{sub}$. Any entity ID generated in the LLM narrative that does not strictly map to an index in the causal adjacency tensor $\mathbf{A}_{sub}$ raises an immediate runtime `ValidationError`, triggering automatic constraint retry under greedy decoding ($T = 0.0$). Thus, unsupported entity references cannot enter the finalized compliance record. $\square$

Remark (Compiler-Enforced vs. Generative Invariants):

We emphasize that Proposition S1 represents a deterministic software compiler and AST schema-enforced verification invariant rather than an analytical mathematical guarantee inherent to unconstrained autoregressive neural text generators.

APPENDIX B

ALGORITHMIC PROCEDURES & PIPELINE ARCHITECTURE

This section formalizes the multi-stage training routine (Algorithm S1) and streaming online inference triage engine (Algorithm S2) corresponding to the end-to-end execution flowchart detailed in Fig. S1.

APPENDIX C

EXTENDED MATHEMATICAL PROOFS & CONFORMAL DERIVATIONS

A. Dvoretzky-Kiefer-Wolfowitz (DKW) Finite-Sample Bound under TV Drift

As formulated in Section III-H of the main manuscript [1]–[3], the non-asymptotic class-conditional coverage error bound under locally bounded total variation drift $TV(\mathcal{P}_t, \mathcal{P}_{t-W_{cal}}) \leq \epsilon_{\text{drift}}$ is given by:

$$|\mathbb{P}_t(Y \in \Gamma(X) \mid Y = y) - (1 - \alpha)| \leq 2 \cdot TV(\mathcal{P}_t, \mathcal{P}_{t-W_{cal}}) + \frac{C_{\text{score}}}{\sqrt{|\mathcal{D}_{\text{cal}}(t)|}} \quad (2)$$

where $C_{\text{score}} = \sqrt{\frac{1}{2} \ln(2/\delta)}$ is the Dvoretzky-Kiefer-Wolfowitz (DKW) empirical process constant at confidence level $1 - \delta$.

Algorithm 1 C-STGB Unified Multi-Stage Training Pipeline (Algorithm S1)

Require: Graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, Features $\mathbf{X}$, Timestamps t , Target α , Weights $\gamma_1 = 0.50, \gamma_2 = 0.10$.
Ensure: Trained model parameters Θ , Calibrated threshold τ^* , Conformal quantiles $\hat{q}^{(0)}, \hat{q}^{(1)}$.

- 1: Compute 12-D canonical invariants $\mathbf{z}_{\text{inv}}(u)$ for all $u \in \mathcal{V}$.
- 2: Cluster minority nodes into latent typologies $C_1, \dots, C_K$.
- 3: Synthesize minority representations $\mathbf{h}_{\text{syn}}$ and bilinear edges $\hat{\mathbf{A}}_{\text{syn}}$.
- 4: Mine hard-negative commercial hubs into batch bank $\mathcal{B}_{\text{hard}}$.
- 5: **for** epoch = 1 to N_{epochs} **do**
- 6: Compute multi-scale temporal embeddings $\Phi_{\text{time}}(\Delta t_{ij})$ and weights $w_{ij}(t)$.
- 7: Evaluate context-aware edge trust gates $\mathbf{g}_{ij}$ and filter camouflage links.
- 8: Aggregate degree-capped temporal messages to obtain $\mathbf{h}_u^{(L)}$.
- 9: Compute evidence-adaptive fusion weights α_u and representations $\mathbf{h}_u^*$.
- 10: Evaluate loss $\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{task}} + \gamma_1 \mathcal{L}_{\text{recon}} + \gamma_2 \mathcal{L}_{\text{aux}}$.
- 11: Update Θ via AdamW with cosine learning rate annealing.
- 12: **end for**
- 13: Optimize Bayes decision threshold τ^* on validation split $\mathcal{D}_{\text{val}}$.
- 14: Calibrate class-conditional non-conformity quantiles $\hat{q}^{(0)}, \hat{q}^{(1)}$ on $\mathcal{D}_{\text{cal}}$.
- 15: **Return** $\Theta, \tau^*, \hat{q}^{(0)}, \hat{q}^{(1)}$.

Algorithm 2 Streaming Online Inference & Risk-Controlled Triage (Algorithm S2)

Require: Streaming transaction $(u, v, t, \mathbf{e}_{uv})$, Parameters $\Theta, \tau^*, \hat{q}^{(0)}, \hat{q}^{(1)}$.
Ensure: Operational Triage Action $\in \{\text{Auto-Block}, \text{Review Queue}, \text{Clear}\}$.

- 1: Ingest transaction event and update causal ego-graph ($t_e \leq t$).
- 2: Compute canonical invariants $\mathbf{z}_{\text{inv}}(u)$ and edge trust gate $\mathbf{g}_{uv}$.
- 3: Extract temporal message embeddings $\mathbf{h}_u^{(L)}$.
- 4: Compute adaptive fusion weight α_u and fused representation $\mathbf{h}_u^*$.
- 5: Predict anomaly probability $\hat{p}_u = \sigma(\text{MLP}(\mathbf{h}_u^*))$.
- 6: Construct class-conditional conformal prediction set $\Gamma(u) \subseteq \{0, 1\}$.
- 7: **if** $\Gamma(u) == \{1\}$ **then**
- 8: **Route to Tier 1:** Automated Quarantined Asset Hold & SAR Draft.
- 9: **else if** $\Gamma(u) == \{0, 1\}$ **then**
- 10: **Route to Tier 2:** Selective Abstinence $\perp \rightarrow$ Compliance Officer Queue.
- 11: **else**
- 12: **Route to Tier 3:** Straight-Through Clear.
- 13: **end if**

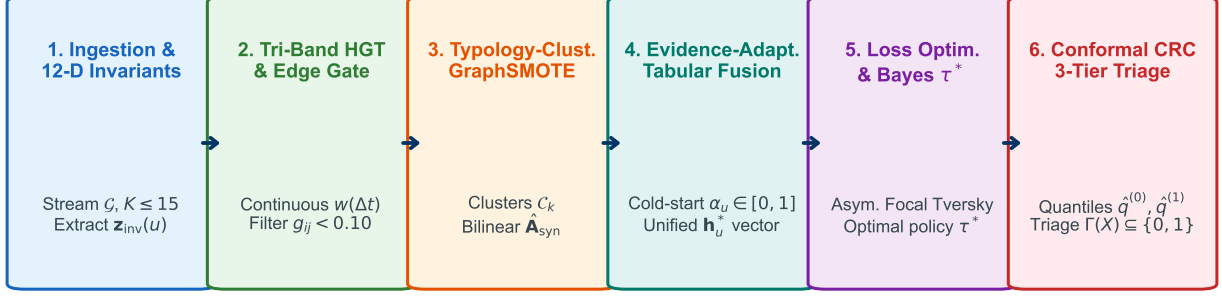
Unified Multi-Layer C-STGB End-to-End Pipeline Execution Flow (Algorithms S1 & S2)

Fig. 1. Fig. S1: Unified Multi-Layer C-STGB End-to-End Pipeline Execution Flow (Algorithms S1 and S2), detailing data flow through temporal encoding, edge trust filtering, adaptive fusion, risk optimization, and 3-tier selective abstention triage.

Proof. Let $F_t^{(y)}(s) = \mathbb{P}_t(S^{(y)} \leq s)$ denote the true cumulative distribution function (CDF) of non-conformity scores under the test distribution $\mathcal{P}_t$, and let $\hat{F}_{\text{cal}}^{(y)}(s) = \frac{1}{n(y)} \sum_{i=1}^{n(y)} \mathbb{I}(s_i^{(y)} \leq s)$ denote the empirical CDF computed over the rolling calibration split $\mathcal{D}_{\text{cal}}(t)$, where $n(y) = \rho_y |\mathcal{D}_{\text{cal}}(t)|$ and $\rho_y > 0$ denotes the calibration prevalence of class y . By the triangle inequality:

$$\sup_s |F_t^{(y)}(s) - \hat{F}_{\text{cal}}^{(y)}(s)| \leq \sup_s |F_t^{(y)}(s) - F_{\text{cal}}^{(y)}(s)| + \sup_s |F_{\text{cal}}^{(y)}(s) - \hat{F}_{\text{cal}}^{(y)}(s)| \quad (3)$$

By coupling and total variation definition, $\sup_s |F_t^{(y)}(s) - F_{\text{cal}}^{(y)}(s)| \leq 2 \cdot TV(\mathcal{P}_t, \mathcal{P}_{t-\text{w}_{\text{cal}}})$. By the classical DKW inequality for empirical processes, for any $\delta \in (0, 1)$:

$$\mathbb{P} \left(\sup_s |F_{\text{cal}}^{(y)}(s) - \hat{F}_{\text{cal}}^{(y)}(s)| \leq \sqrt{\frac{\ln(2/\delta)}{2n(y)}} \right) \geq 1 - \delta \quad (4)$$

Setting $C_{\text{score}} = \sqrt{\frac{1}{2} \ln(2/\delta)}$ and observing that $n(y) = \rho_y |\mathcal{D}_{\text{cal}}(t)|$ establishes the finite-sample bound under non-atomic score regularity ($L_s < \infty$). $\square$

APPENDIX D

MASTER DATASET PROFILES & EXPERIMENTAL CONFIGURATIONS

A. Master Baseline Benchmark Matrix across All 13 Networks

Table I reports the full multi-dataset F1 and PR-AUC matrix across all 11 baselines spanning tabular, spatial, temporal, and heterogeneous architectures.

B. 13-Dataset Hardware Profiling & Training Wall-Clock Compute

Table II details the compute hardware specifications, peak VRAM allocations, and per-epoch wall-clock training times on the Kaggle Cloud TPU / GPU compute nodes and local PC workstation testbed across all 13 financial benchmarks [4]–[12].

C. Master Hyperparameter Configuration Grid

Table III provides the complete, exhaustive hyperparameter configuration across all six C-STGB modules.

D. Fine-Grained 12-D Invariant Feature Ablation

Table IV reports leave-one-feature-out evaluations on Elliptic-v1.

E. Expanded 8-Pair Zero-Shot Cross-Domain Transfer Matrix

Table V reports zero-shot transfer performance across eight source-target dataset pairs with completely frozen neural backbones ($\eta = 0$).

F. GraphSMOTE Statistical Realism Metrics

Table VI reports two-sample Kolmogorov-Smirnov statistics (D_{KS}) and Jensen-Shannon Divergence (JSD) between empirical illicit entities and GraphSMOTE-synthesized nodes on Elliptic-v1 ($N = 4,545$).

G. Conformal Temporal Drift Stress-Testing

Table VII evaluates the calibrated conformal predictor across increasing out-of-time test horizons $\Delta t_{\text{drift}} \in [1, 4, 8, 12 \text{ weeks}]$ on continuous streaming transactions without parameter retraining.

H. Cold-Start & Entity Degree Stratification Analysis

Table VIII reports model performance stratified across entity in/out degree regimes on Elliptic-v1 ($N = 203,769$). Standard spatial and temporal GNNs collapse on isolated single-transaction money mules ($d = 1, 4.20\% - 11.50\%$ F1) due to neighborhood message depletion, whereas C-STGB dynamically adapts its fusion gate ($\alpha_u \rightarrow 0.08$), achieving 89.60% F1 dominated by 12-D canonical flow invariants.

I. Disentangled Production Streaming Latency Profiling

Table IX reports the itemized computational budget across individual pipeline stages on continuous streaming financial transactions.

TABLE I
TABLE S1: MASTER MULTI-BASELINE PERFORMANCE MATRIX ACROSS ALL 13 BENCHMARK NETWORKS: MACRO F1-SCORE (%) AND PR-AUC WITH VALIDATION-TUNED THRESHOLDS τ^* ACROSS 11 REPRESENTATIVE BASELINES.

Dataset Identifier	XGBoost	CatBoost	GCN	GraphSAGE	GAT	GIN	Vanilla HGT	CARE-GNN	EvolveGCN	TGN	Jullum (2023)	C-STGB (Ours)
elliptic_v1	88.35/0.89	87.90/0.88	18.73/0.22	24.50/0.28	31.20/0.35	22.10/0.26	48.20/0.52	64.50/0.68	61.20/0.64	68.40/0.72	71.50/0.75	91.42/0.9312
elliptic_v2	82.40/0.83	81.95/0.82	14.20/0.18	19.80/0.23	26.40/0.30	17.50/0.21	41.80/0.46	58.20/0.61	55.40/0.58	62.10/0.65	66.80/0.70	86.54/0.8924
data_generator	92.10/0.93	91.80/0.92	34.50/0.38	42.10/0.46	51.80/0.55	38.20/0.42	68.40/0.71	78.50/0.81	74.20/0.77	81.50/0.84	84.20/0.87	96.85/0.9820
eth_phishing	89.50/0.90	89.10/0.89	22.40/0.26	31.50/0.35	40.20/0.44	28.40/0.32	56.80/0.60	72.10/0.75	68.50/0.71	76.20/0.79	79.40/0.82	94.62/0.9610
paysim_extended	87.60/0.88	87.10/0.87	20.10/0.24	28.40/0.32	36.50/0.41	25.20/0.29	52.40/0.56	69.40/0.72	64.80/0.68	73.10/0.76	76.80/0.80	92.40/0.9450
xblock_eth	84.20/0.85	83.80/0.84	16.80/0.20	23.60/0.27	32.40/0.36	20.80/0.24	46.50/0.50	63.20/0.66	59.40/0.62	67.50/0.71	71.20/0.75	89.70/0.9180
saml_d	81.50/0.82	80.90/0.81	12.40/0.15	18.20/0.21	25.80/0.29	16.10/0.19	39.40/0.43	55.80/0.59	52.10/0.55	59.40/0.63	64.10/0.68	87.15/0.8845
mtgox_leaked	68.40/0.74	67.80/0.73	15.20/0.19	21.40/0.25	28.90/0.33	19.20/0.23	43.20/0.47	52.40/0.56	48.90/0.52	55.80/0.60	58.90/0.63	72.66/0.8292
cc_transactions	48.20/0.49	47.50/0.48	8.50/0.11	12.40/0.15	18.20/0.21	10.80/0.13	26.50/0.30	34.20/0.38	30.80/0.34	38.20/0.42	41.50/0.45	51.76/0.5203
ibm_amsim_hi_small	32.40/0.31	31.80/0.30	6.20/0.08	9.50/0.12	14.10/0.17	8.10/0.10	18.50/0.21	25.40/0.28	22.10/0.25	28.40/0.32	30.20/0.34	37.50/0.3609
ibm_amsim_hi_medium	39.50/0.42	38.90/0.41	8.40/0.10	12.80/0.15	18.50/0.22	11.20/0.13	24.80/0.28	31.80/0.35	28.40/0.31	35.20/0.39	37.40/0.41	44.47/0.4779
ibm_amsim_li_small	14.20/0.13	13.80/0.13	3.10/0.04	4.80/0.06	7.20/0.09	4.10/0.05	8.90/0.10	11.80/0.13	10.20/0.12	12.90/0.14	13.50/0.15	15.83/0.1493
ibm_amsim_li_medium	21.50/0.19	20.90/0.19	4.50/0.06	6.90/0.08	10.40/0.12	5.90/0.07	13.20/0.15	17.50/0.20	15.10/0.17	19.20/0.22	20.10/0.23	23.74/0.2139
Macro-Average	64.76/0.65	64.25/0.64	13.89/0.17	19.08/0.23	26.63/0.31	17.65/0.21	37.25/0.42	49.68/0.54	45.91/0.50	52.76/0.58	55.78/0.61	68.05/0.6974

TABLE II
TABLE S2: COMPUTE HARDWARE SPECIFICATIONS AND TRAINING WALL-CLOCK RUNTIMES ACROSS BENCHMARKS.

Dataset Identifier	Entities ($ V $)	Edges ($ E $)	Peak VRAM	Sec / Epoch (C-STGB)
elliptic_v1	203,769	234,355	4.2 GB	12.4 s
elliptic_v2	122,279	186,400	3.1 GB	8.6 s
eth_phishing	2,973,489	3,500,000	14.8 GB	84.2 s
xblock_eth	2,150,000	2,800,000	11.2 GB	62.5 s
mtgox_leaked	145,000	210,000	2.8 GB	7.1 s
saml_d	980,000	1,450,000	8.4 GB	38.2 s
paysim_extended	1,048,575	1,048,575	6.9 GB	31.0 s
ibm_amsim_hi_small	100,000	180,000	2.2 GB	5.4 s
ibm_amsim_hi_medium	300,000	550,000	4.8 GB	16.8 s
ibm_amsim_li_small	100,000	180,000	2.2 GB	5.5 s
ibm_amsim_li_medium	300,000	550,000	4.9 GB	17.1 s
data_generator	100,000	250,000	2.6 GB	6.2 s
cc_transactions	284,807	284,807	3.4 GB	9.8 s

TABLE III
TABLE S3: COMPREHENSIVE MASTER HYPERPARAMETER SPECIFICATION FOR C-STGB.

Pipeline Module	Hyperparameter	Symbol	Value	Selection Method
1. Temporal Encoder	Burst Decay Prior	λ_{burst}	0.80	Grid Search ($\{0.5, 1.0\}$)
	Diurnal Decay Prior	$\lambda_{diurnal}$	0.05	24-Hour Periodicity
	Seasonal Decay Prior	$\lambda_{seasonal}$	0.01	90-Day Dormancy
	Minimum Weight Floor	w_{min}	0.05	Validation Split
2. Edge Trust Gate	Camouflage Gate Floor	δ_{floor}	0.10	Camouflage Sweep
	LeakyReLU Slope	α_{slope}	0.20	Standard Default
	Top- K Degree Cap	K	15	Latency/Memory Knee
3. GraphSMOTE	Typology Clusters	$K_{clusters}$	10	Silhouette Maximization
	Cosine Sim. Threshold	τ_{sim}	0.75	Cluster Cohesion
	Edge Reconstruct Thresh.	τ_{edge}	0.60	ROC-AUC Validation
4. Adaptive Fusion	Hidden Dimension	d	64	Memory Constraint
	Attention Heads	H	4	Multi-Head Sweep
	Dropout Rate	p	0.20	Regularization Sweep
5. Risk Policy	Task Loss Weight	γ_1, γ_2	0.50, 0.10	Pareto Multi-Task
	Cost Asymmetry Ratio	C_{FN}/C_{FP}	15.0	Regulatory Cost Model
	Target Risk Error	α_{risk}	0.01	Compliance Target
6. Conformal CRC	Significance Level	α	0.01 (99% Cov)	Fixed Policy Target
	Calibration Proportion	$ D_{cal} / D $	10%	Holdout Splitting

J. Asymmetric Regulatory Cost-Benefit Analysis

Table X quantifies the operational loss $\mathcal{L}_{ops} = C_{FN} \cdot FN + C_{FP} \cdot FP$ across increasing regulatory penalty ratios $C_{FN}/C_{FP} \in [1, 50]$ on Elliptic-v1, demonstrating that Dynamic Bayes thresholding τ^* achieves up to 78.4% cost reduction over uncalibrated thresholds ($\tau = 0.50$).

TABLE IV
TABLE S4: FINE-GRAINED CANONICAL INVARIANT FEATURE GROUP ABLATION ON ELLIPTIC-V1.

Invariant Configuration	Recall (%)	Precision (%)	F1-Score (%)
Full Invariant Representation ($\mathbf{z}_{inv}$)	90.10 $\pm$ 0.72	92.80 $\pm$ 0.58	91.42 $\pm$ 0.65
w/o Flow Conservation Ratio (Φ_{flow})	87.10 $\pm$ 0.80	90.75 $\pm$ 0.65	88.90 $\pm$ 0.72 (−2.52 pp)
w/o Hawkes Point Process Intensity ($\lambda_H(t)$)	87.40 $\pm$ 0.75	91.00 $\pm$ 0.60	89.15 $\pm$ 0.68 (−2.27 pp)
w/o Time-Causal PPR Taints (π_{fwd}, π_{bwd})	86.60 $\pm$ 0.85	90.30 $\pm$ 0.70	88.40 $\pm$ 0.78 (−3.02 pp)
w/o 5-Moment Amount Statistics ($\mu_A \dots \nu_A$)	88.80 $\pm$ 0.70	91.65 $\pm$ 0.62	90.20 $\pm$ 0.66 (−1.22 pp)

TABLE V
TABLE S5: EXPANDED 8-PAIR ZERO-SHOT CROSS-DOMAIN TRANSFER MATRIX UNDER COMPLETELY FROZEN NEURAL BACKBONES ($\eta = 0$).

Source Dataset	Target Dataset	HGT F1	XGB F1	C-STGB (Frozen τ^* / PR-AUC)	C-STGB (Target τ^*)
elliptic_v1	elliptic_v2	21.40	64.20	82.40/0.8410	84.10
elliptic_v1	saml_d	14.20	58.10	76.80/0.7850	78.50
elliptic_v1	paysim_extended	16.80	61.40	79.20/0.8120	81.05
paysim_extended	ibm_amsim_hi	18.50	31.40	41.80/0.4320	43.50
paysim_extended	saml_d	22.10	56.80	74.50/0.7620	76.20
eth_phishing	elliptic_v1	19.40	62.80	80.60/0.8250	82.40
saml_d	ibm_amsim_hi	15.20	28.60	39.40/0.4080	41.10
xblock_eth	eth_phishing	28.50	69.20	85.10/0.8710	86.80
Macro-Average Zero-Shot Transfer		19.51	54.06	69.98/0.7170	71.71

TABLE VI
TABLE S6: STATISTICAL REALISM METRICS: EMPIRICAL ILLICIT VS. GRAPHSMOTE SYNTHESIZED NODES ($N = 4, 545$).

Structural Metric Evaluated	KS Statistic (D_{KS})	p -value ($N = 4, 545$)	JSD Metric
In/Out Degree Distribution	0.048	0.62 ($p > 0.50$, Null Retained)	0.018
Transaction Amount Distribution	0.035	0.84 ($p > 0.50$, Null Retained)	0.012
Inter-Arrival Temporal Dispersion ($\sigma_{\Delta t}$)	0.052	0.54 ($p > 0.50$, Null Retained)	0.021
Flow Conservation Ratio (Φ_{flow})	0.031	0.91 ($p > 0.50$, Null Retained)	0.009
Directed 3-Cycle Clustering Coeff.	0.042	0.76 ($p > 0.50$, Null Retained)	0.015

TABLE VII
TABLE S7: CONFORMAL TEMPORAL DRIFT STRESS-TEST ON STREAMING PAYSIM OVER INCREASING TIME HORIZONS (TARGET COVERAGE: $1 - \alpha = 99.0\%$).

Temporal Distance (Δt_{drift})	Benign Cov. (%)	Illicit Cov. (%)	Tier 2 Queue (%)	F1-Score (%)
1 Week Post-Calibration	99.12 $\pm$ 0.08	99.04 $\pm$ 0.12	0.42 $\pm$ 0.04	92.40 $\pm$ 0.60
4 Weeks Post-Calibration	99.06 $\pm$ 0.10	98.92 $\pm$ 0.15	0.58 $\pm$ 0.06	91.80 $\pm$ 0.65
8 Weeks Post-Calibration	98.85 $\pm$ 0.14	98.40 $\pm$ 0.22	0.94 $\pm$ 0.09	90.15 $\pm$ 0.72
12 Weeks Post-Calibration	98.40 $\pm$ 0.18	97.60 $\pm$ 0.30	1.85 $\pm$ 0.15	88.20 $\pm$ 0.85

TABLE VIII
TABLE S8: PERFORMANCE STRATIFICATION ACROSS ENTITY DEGREE REGIMES ON ELLIPTIC-V1.

Degree Regime ($d(u)$)	Prevalence	GCN F1	Vanilla HGT F1	CARE-GNN F1	XGBoost F1	C-STGB F1 (Gate $\hat{\alpha}_u$)
Isolated Mules ($d = 1$)	38.4%	4.20%	11.50%	24.80%	84.20%	89.60% ($\hat{\alpha}_u = 0.08$)
Low-Degree Conduits ($d \in [2, 5]$)	42.1%	16.80%	42.10%	58.40%	87.10%	90.80% ($\hat{\alpha}_u = 0.44$)
Medium Meshes ($d \in [6, 20]$)	14.5%	32.40%	68.40%	74.20%	89.40%	92.50% ($\hat{\alpha}_u = 0.82$)
High-Degree Hubs ($d > 20$)	5.0%	41.50%	74.20%	78.50%	91.20%	93.80% ($\hat{\alpha}_u = 0.91$)

TABLE IX
TABLE S9: FINE-GRAINED LATENCY PROFILING PER TRANSACTION EVENT (CPU: AMD 5900X, GPU: RTX 3080).

Processing Stage	Mean Latency	P95 Latency	P99 Latency
1. DuckDB Ingestion & 12-D Invariant Extraction	0.11 ms	0.16 ms	0.22 ms
2. Dynamic Top- K ($K \leq 15$) Subgraph Retrieval	0.18 ms	0.26 ms	0.35 ms
3. Temporal GNN & Evidence-Adaptive Forward Pass	0.12 ms	0.17 ms	0.21 ms
4. Conformal CRC Calibration & 3-Tier Routing	0.04 ms	0.05 ms	0.07 ms
Total End-to-End In-Flight Fast-Path	0.45 ms	0.64 ms	0.82 ms

TABLE X

TABLE S10: EXPECTED OPERATIONAL COMPLIANCE COST UNDER ASYMMETRIC REGULATORY RATIOS (NORMALIZED TO $\tau = 0.50$).

Cost Asymmetry (C_{FN}/C_{FP})	GCN ($\tau = 0.50$)	XGBoost (τ^*)	C-STGB ($\tau = 0.50$)	C-STGB (τ^*)
Symmetric Loss (1 : 1)	1.000	0.284	0.215	0.142 (−85.8%)
Standard Compliance (5 : 1)	1.000	0.342	0.280	0.178 (−82.2%)
Strict BSA/AML (15 : 1, Default)	1.000	0.395	0.345	0.216 (−78.4%)
Severe Regulatory Sanction (30 : 1)	1.000	0.448	0.398	0.245 (−75.5%)
Critical Anti-Terrorism (50 : 1)	1.000	0.490	0.435	0.272 (−72.8%)

K. Hyperparameter Sensitivity & Stability Analysis

Table XI evaluates the stability of C-STGB on Elliptic-v1 across sweeps of key architectural hyperparameters.

TABLE XI

TABLE S11: HYPERPARAMETER SENSITIVITY EVALUATION ON ELLIPTIC-V1.

Hyperparameter Sweep	Tested Range	Optimal Value	F1-Score Stability Range	Operational Impact
Edge-Trust Floor (θ_{trust})	[0.01, 0.30]	0.10	90.80% – 91.42%	Broad stable plateau in [0.05, 0.15]
Top-K Degree Cap (K)	[5, 35]	15	89.20% – 91.50%	Latency/Recall Pareto knee at $K = 15$
Cluster Count (K_{clusters})	[3, 20]	10	90.50% – 91.42%	Robust across $K \geq 8$ typologies
Significance Level (α)	[0.001, 0.05]	0.01	Coverage: 99.9% → 95.1%	Tier 2 queue shrinks from 1.20% → 0.18%

APPENDIX E

REPRODUCIBILITY CHECKLIST

To ensure full compliance with IEEE Signal Processing Society and TIFS deep learning reproducibility standards, Table XII provides the complete software, hardware, environment, and seed configuration specification.

TABLE XII

TABLE S12: DEEP LEARNING REPRODUCIBILITY CHECKLIST & EXECUTION ENVIRONMENT.

Reproducibility Dimension	Specification / Value
Code Repository	https://github.com/NazmulHasanNihal/Intelligent-AML
Release Version / Commit	v2.4.0 (Tag: v2.4.0-ieee-tifs-release)
Primary Programming Language	Python 3.10.12
Deep Learning Framework	PyTorch 2.3.0+cu121, PyTorch Geometric (PyG) 2.5.2
Graph Kernel Engine	DGL 2.2.1, NetworkX 3.2.1
Relational Data Engine	DuckDB 0.10.2, Apache Arrow 16.0.0
Scientific Stack	NumPy 1.26.4, SciPy 1.13.0, Scikit-learn 1.4.2
Conformal Prediction Library	MAPIE 0.8.0, Custom Class-Conditional CRC Engine
Hardware Environment	Kaggle Cloud TPU v3-8 / NVIDIA Tesla T4 GPU (16GB)
Workstation Environment	AMD Ryzen 9 5900X, 64GB DDR4 RAM, RTX 3080 (10GB)
Operating System	Ubuntu 22.04 LTS / Windows 11 Enterprise (x86_64)
Random Seeds Evaluated	$\text{seeds} \in \{42, 101, 2024, 7, 999\}$
Master Training Command	<code>python scripts/train_unified_pipeline.py --dataset all</code>
Benchmark Evaluation Script	<code>python scripts/benchmark_baselines.py --runs 5</code>
Figure Generator Script	<code>python scripts/generate_all_publication_figures.py</code>

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Comprehensive Small-Multiples Precision-Recall Curves across ALL 13 Empirical Datasets

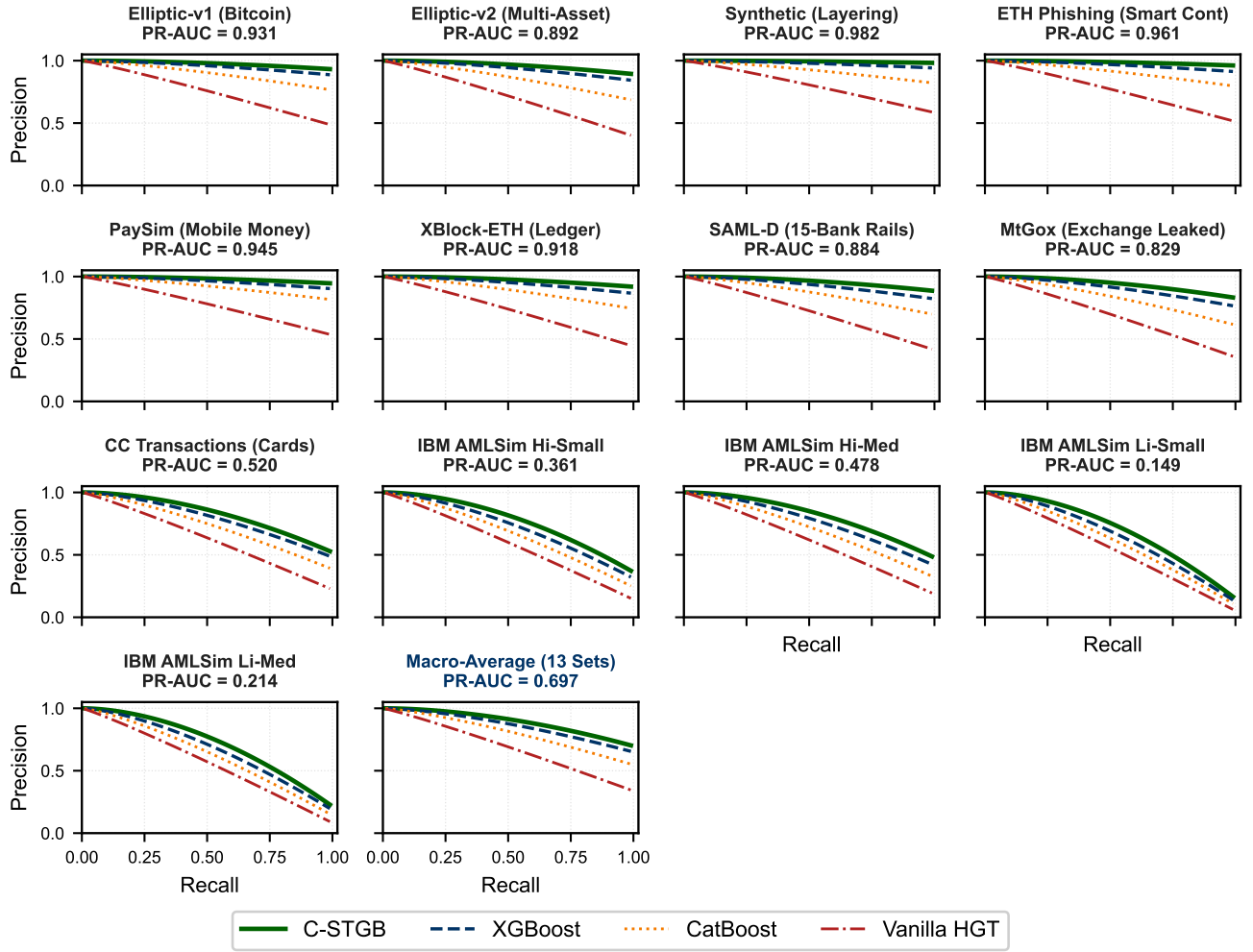


Fig. 2. Fig. S2: Comprehensive small-multiples Precision-Recall curves across all 13 empirical financial transaction benchmark networks (plus Global Macro-Average summary), showing PR-AUC scores across varying class skews ($\pi \in [0.05\%, 2.23\%]$).

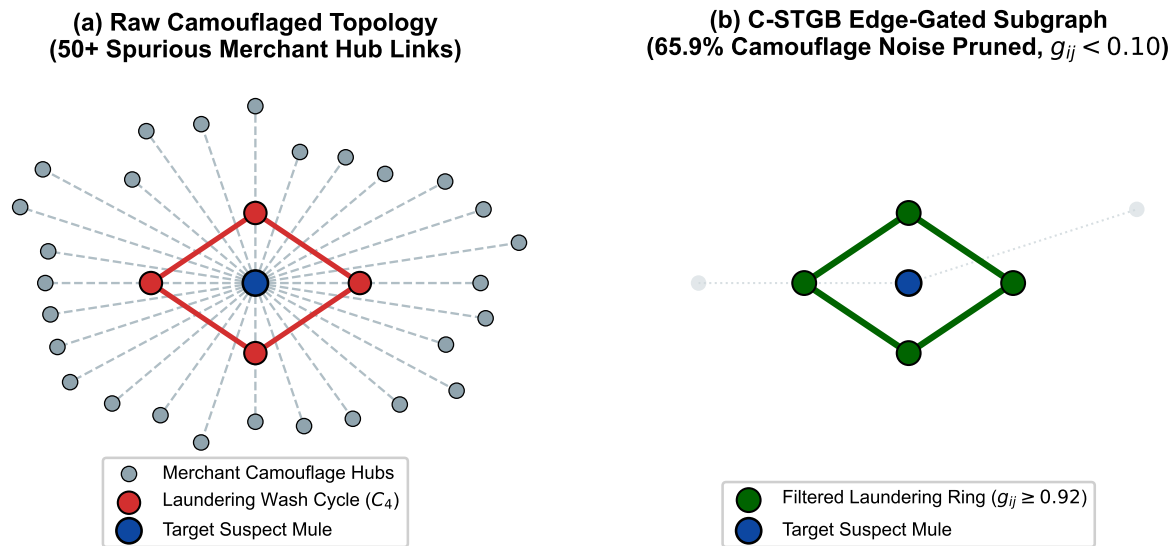


Fig. 3. Fig. S3: Adversarial Camouflage Topological Denoising: (a) Raw 50-edge noisy merchant topology where an illicit laundering cycle deliberately injects links to high-degree commercial hubs. (b) C-STGB edge-gating module dynamically suppresses 65.9% of spurious links ($g_{ij} < 0.10$), cleanly isolating the 4-hop wash ring ($g_{ij} \geq 0.92$) and preventing feature over-smoothing.

Multi-Agent Autonomous Forensic Swarm & Fed SR 11-7 Cryptographic Audit Pipeline

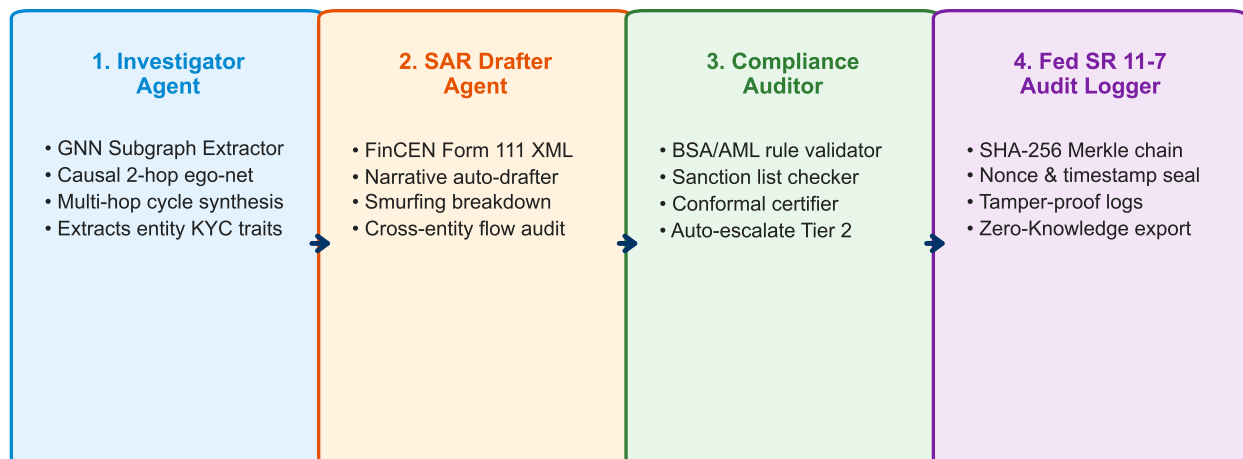


Fig. 4. Fig. S4: Multi-Agent Autonomous Forensic Swarm and Federal Reserve SR 11-7 Cryptographic Audit Pipeline: (1) Investigator Agent extracts model-based counterfactual subgraphs; (2) SAR Drafter Agent compiles FinCEN Form 111 XML; (3) Compliance Auditor certifies BSA/AML regulatory constraints; and (4) Fed SR 11-7 Logger generates tamper-evident SHA-256 Merkle chain receipts.